Code No: 117BX **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, May/June - 2019
COMPUTER GRAPHICS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Describe the working of raster scan Systems. [2]
b) How do we represent polygon using polygon table, edge table and vertex table? Explain with an example. [3]
c) Define Shear transformation. [2]
d) What are the limitations of sutherland hodggman algorithm? [3]
e) Define Clipping. [2]
f) What are the properties of Bezier curve? [3]
g) What are the limitations of octree method? [2]
h) Discuss BSPtree methods. [3]
i) What is the use of key frame system? [2]
j) How is animation sequence constructed using kinematic or dynamic descriptions? [3]

PART-B

(50 Marks)

2.a) Consider a raster display system with resolution of 800 by 400. How many pixels could be accessed per second by a display controller that refreshes the screen at the rate of 120 frames per second?
b) Describe briefly Bresenham's circle drawing algorithm with an example. Why do we prefer incremental algorithm over DDA? [5+5]

OR

3.a) Some of the line-generation algorithms will not draw 450 lines properly. Why? Can you suggest modifications that will correct this flaw?
b) Using midpoint ellipse algorithm, generate points on the ellipse with center as origin, major axis 8 units and minor axis 6 units. [5+5]

4.a) Find reflections of point (5, 6) with respect to a line y = x and find the corresponding transformation matrix.
b) Explain the steps involved in Sutherland-Hodge man algorithm. [5+5]

OR

5.a) Explain about the midpoint sub division line clipping algorithm.
b) What is the need of homogeneous coordinates? Give the homogeneous coordinates for translation, rotation and scaling. [5+5]

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6.a) How are periodic B-spline curves different from non-periodic B-spline curves?
b) Given control points (20,100), (50,100), (60,120) and (110,150). Calculate coordinates of any four points lying on the corresponding Beizer curve. [5+5]

OR

7.a) Derive the transformation matrix for Hermite curve.
b) Derive the transformation matrix for rotation about an x-axis in 3D. [5+5]

8.a) Discuss in brief about 3D shear transformations.
b) Derive the matrix form for the rotation about z- axis in 3-D space. [5+5]

OR

9.a) What are the steps involved in depth buffer algorithm?
b) Explain about the Back Face detection algorithm. What are the limitations of the Back Face detection method? [5+5]

10.a) What are key frames? Explain its importance.
b) Explain in brief about computer animation functions. [5+5]

OR

11.a) What is morphing? How is a shape converted into another shape by morphing?
b) Describe the techniques to achieve the simple animation effects. [5+5]

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